

ENV 790.30 - Time Series Analysis for Energy Data | Spring 2025

Assignment 3 - Due date 02/03/26

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Directions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., “LuanaLima_TSA_A03_Sp25.Rmd”). Then change “Student Name” on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

Please keep this R code chunk options for the report. It is easier for us to grade when we can see code and output together. And the tidy.opts will make sure that line breaks on your code chunks are automatically added for better visualization.

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

Questions

Consider the same data you used for A2 from the spreadsheet “Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source.xlsx”. The data comes from the US Energy Information and Administration and corresponds to the December 2025 Monthly Energy Review. This time you will work only with the following columns: **Total Renewable Energy Production**; and **Hydroelectric Power Consumption**.

Create a data frame structure with these two time series only.

R packages needed for this assignment: “forecast”, “tseries”, and “Kendall”. Install these packages, if you haven’t done yet. Do not forget to load them before running your script, since they are NOT default packages.\

```
#Load/install required package here
library("tidyverse")
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2    4.0.1      v tibble     3.2.1
## v lubridate  1.9.3      v tidyr      1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()      masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library("forecast")
```

```
## Registered S3 method overwritten by 'quantmod':
##   method      from
##   as.zoo.data.frame zoo
```

```
library("tseries")
```

```
install.packages("Kendall")
```

```
## Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.4'
## (as 'lib' is unspecified)
```

```
library("Kendall")
```

```
#Read Excel file
library(readxl)
library(openxlsx)

getwd()
```

```
## [1] "/home/guest/TSA_Sp26"
```

```
Renewable_Energy <- read_excel(path = "./Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_Sector",
                               skip = 12, sheet = "Monthly Data", col_names = FALSE)
```

```
## New names:
## * ' ' -> '...1'
## * ' ' -> '...2'
## * ' ' -> '...3'
## * ' ' -> '...4'
## * ' ' -> '...5'
## * ' ' -> '...6'
## * ' ' -> '...7'
## * ' ' -> '...8'
## * ' ' -> '...9'
## * ' ' -> '...10'
## * ' ' -> '...11'
## * ' ' -> '...12'
## * ' ' -> '...13'
## * ' ' -> '...14'
```

```
read_col_names <- read_excel(path = "./Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_Sector",
                              skip = 10, n_max = 1, sheet = "Monthly Data", col_names = FALSE)
```

```
## New names:
## * ' ' -> '...1'
```

```
## * '' -> '...2'
## * '' -> '...3'
## * '' -> '...4'
## * '' -> '...5'
## * '' -> '...6'
## * '' -> '...7'
## * '' -> '...8'
## * '' -> '...9'
## * '' -> '...10'
## * '' -> '...11'
## * '' -> '...12'
## * '' -> '...13'
## * '' -> '...14'
```

```
colnames(Renewable_Energy) <- read_col_names
view(Renewable_Energy)

Renewable_Energy$Month <- as.Date(Renewable_Energy$Month)
class(Renewable_Energy$Month)
```

```
## [1] "Date"
```

```
#Creating data frames
total_renewable_production <- select(Renewable_Energy,
                                     "Month", "Total Renewable Energy Production")
view(total_renewable_production)

hydro_consumption <- select(Renewable_Energy,
                            "Month", "Hydroelectric Power Consumption")
view(hydro_consumption)
```

```
##Trend Component
```

Q1

For each series (Total Renewable Production and Hydroelectric Consumption) create three plots arranged in a row (side-by-side): (1) time series plot, (2) ACF, (3) PACF. Use `cowplot::plot_grid()` to place them in a grid.

```
install.packages("ggplot2")
```

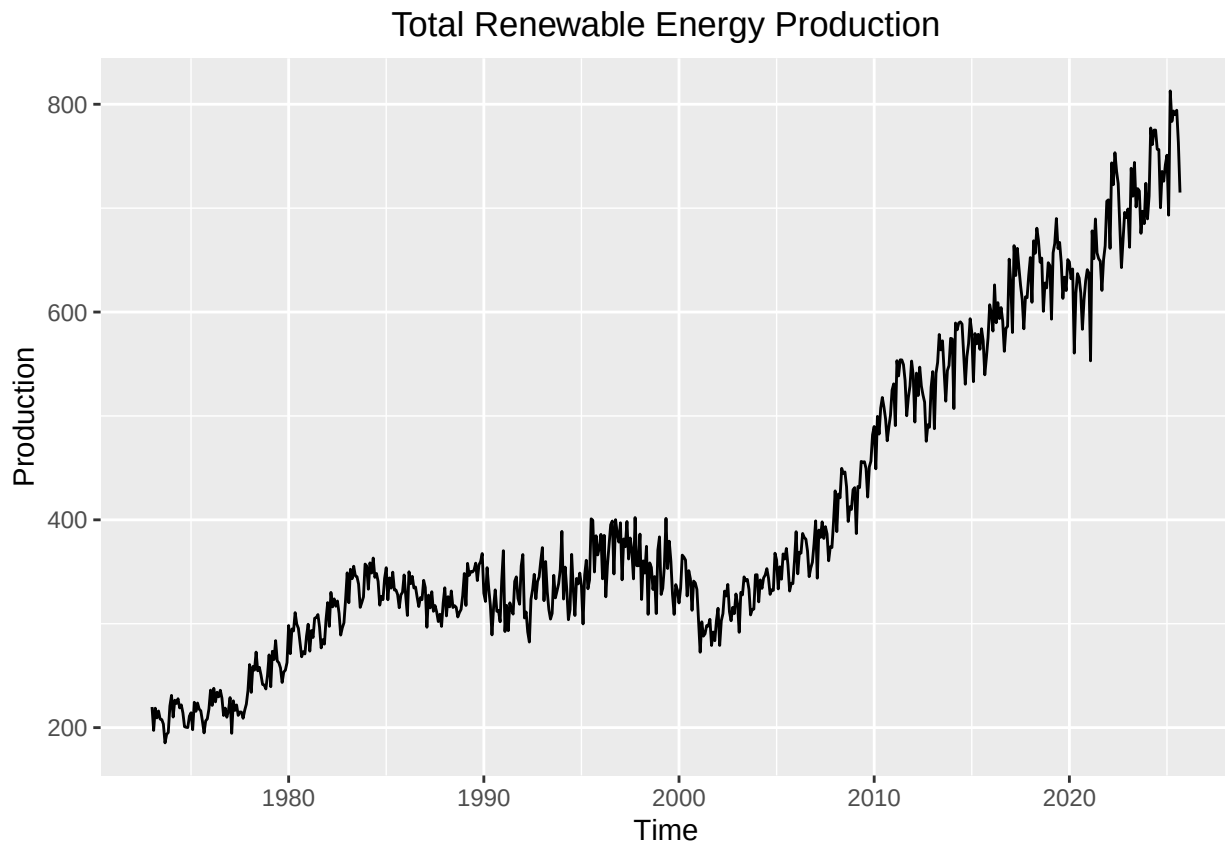
```
## Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.4'
## (as 'lib' is unspecified)
```

```
library("ggplot2")
library(cowplot)
```

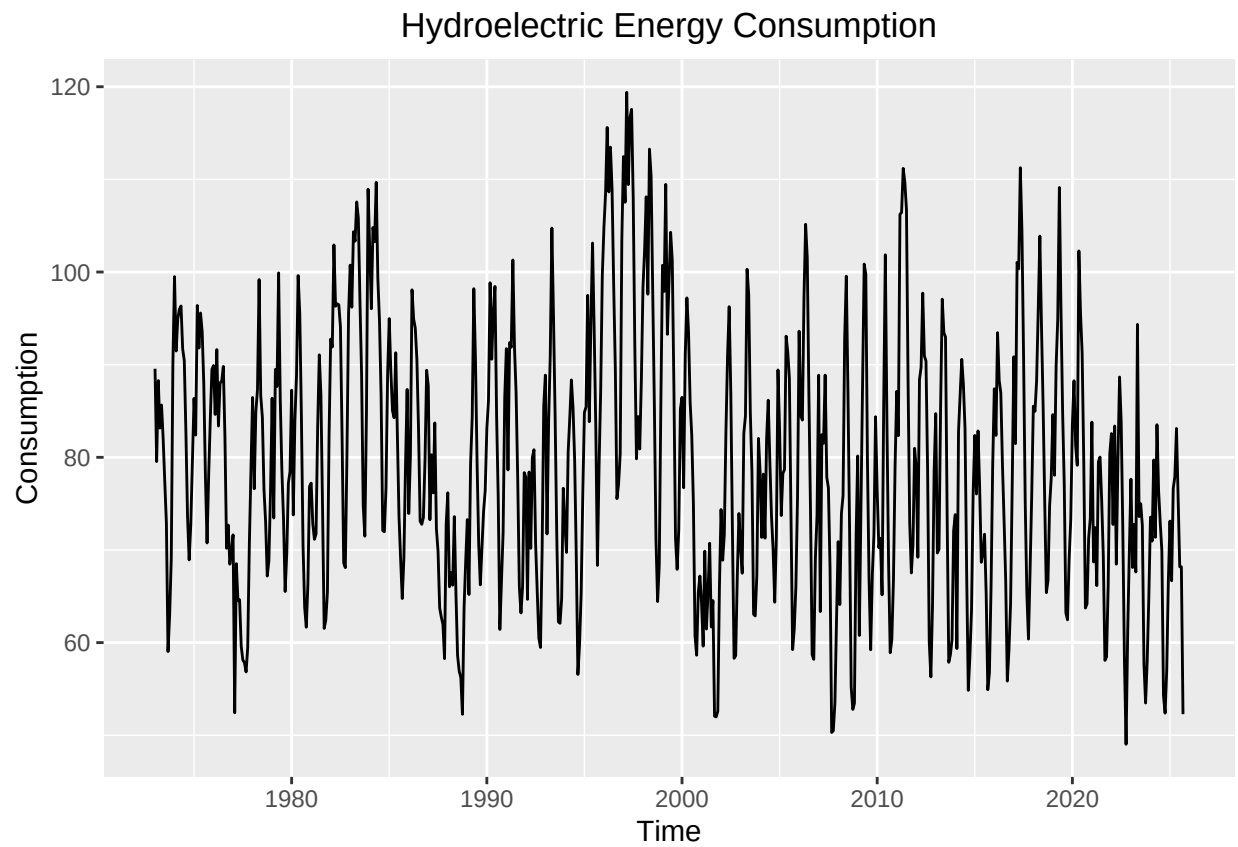
```
##
## Attaching package: 'cowplot'
```

```
## The following object is masked from 'package:lubridate':  
##  
## stamp
```

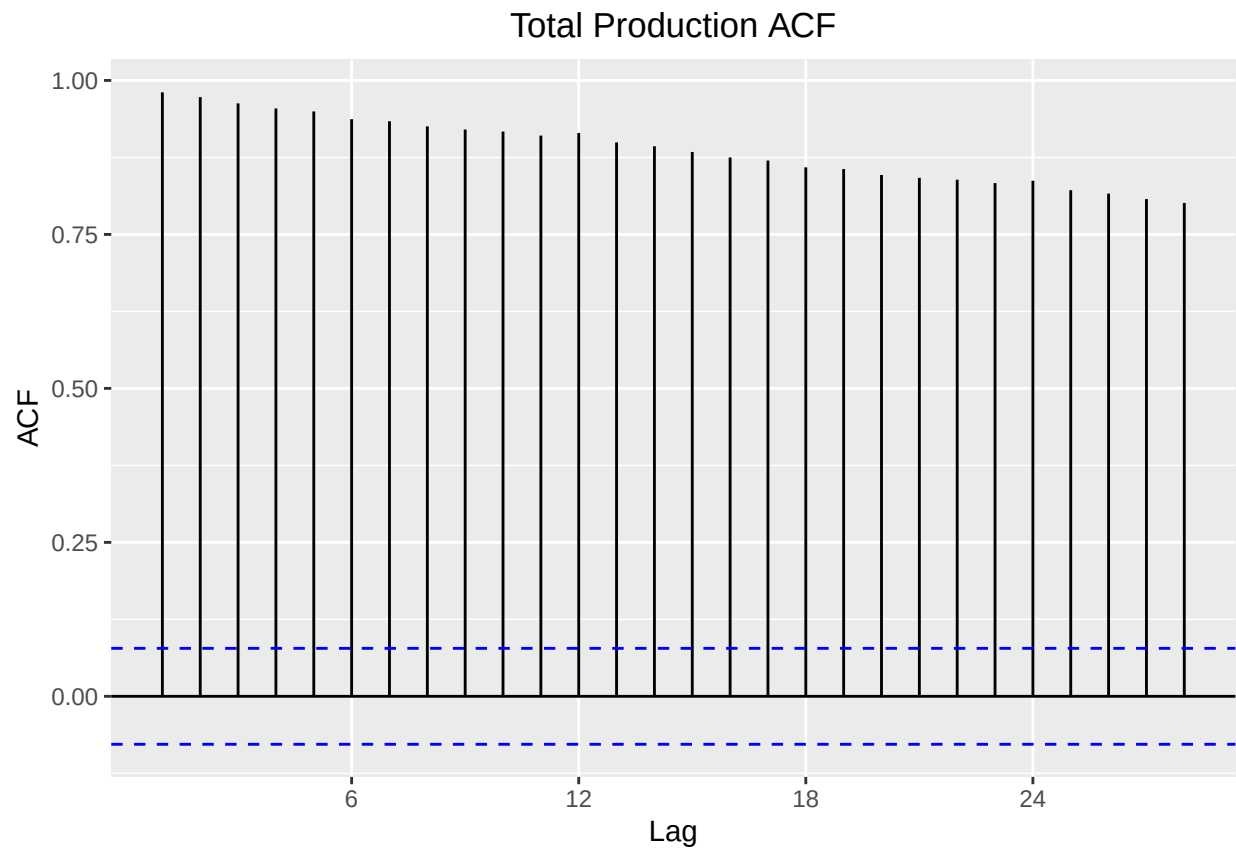
```
#time series  
total_production_ts <- ts(total_renewable_production[,1:2],  
                          start = c(1973,1),  
                          frequency = 12)  
  
hydro_consumption_ts <- ts(hydro_consumption[,1:2],  
                           start = c(1973,1),  
                           frequency = 12)  
  
total_plot <- autoplot(total_production_ts[,2]) +  
  xlab("Time") +  
  ylab("Production") +  
  ggtitle("Total Renewable Energy Production") +  
  theme(plot.title = element_text(hjust = 0.5))  
print(total_plot)
```



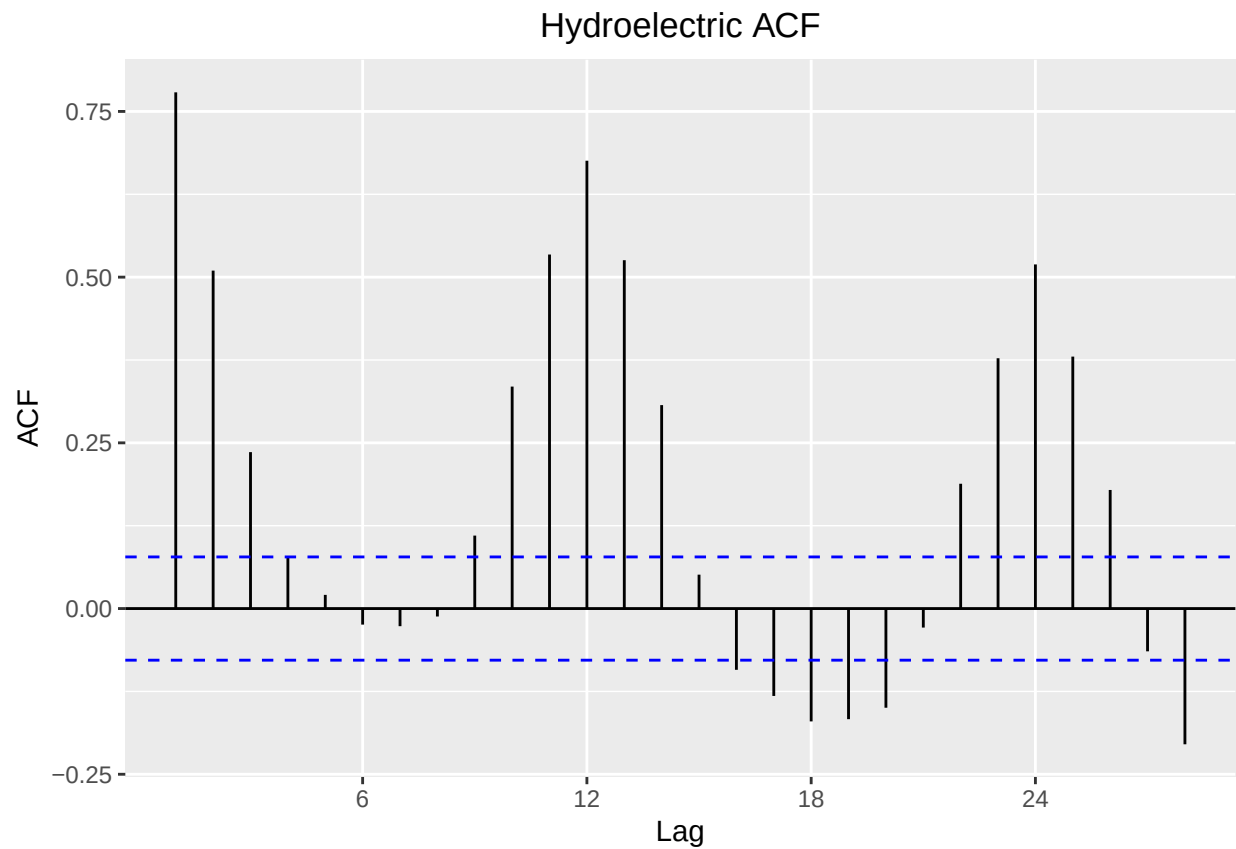
```
hydro_plot <- autoplot(hydro_consumption_ts[,2]) +  
  xlab("Time") +  
  ylab("Consumption") +  
  ggtitle("Hydroelectric Energy Consumption") +  
  theme(plot.title = element_text(hjust = 0.5))  
print(hydro_plot)
```



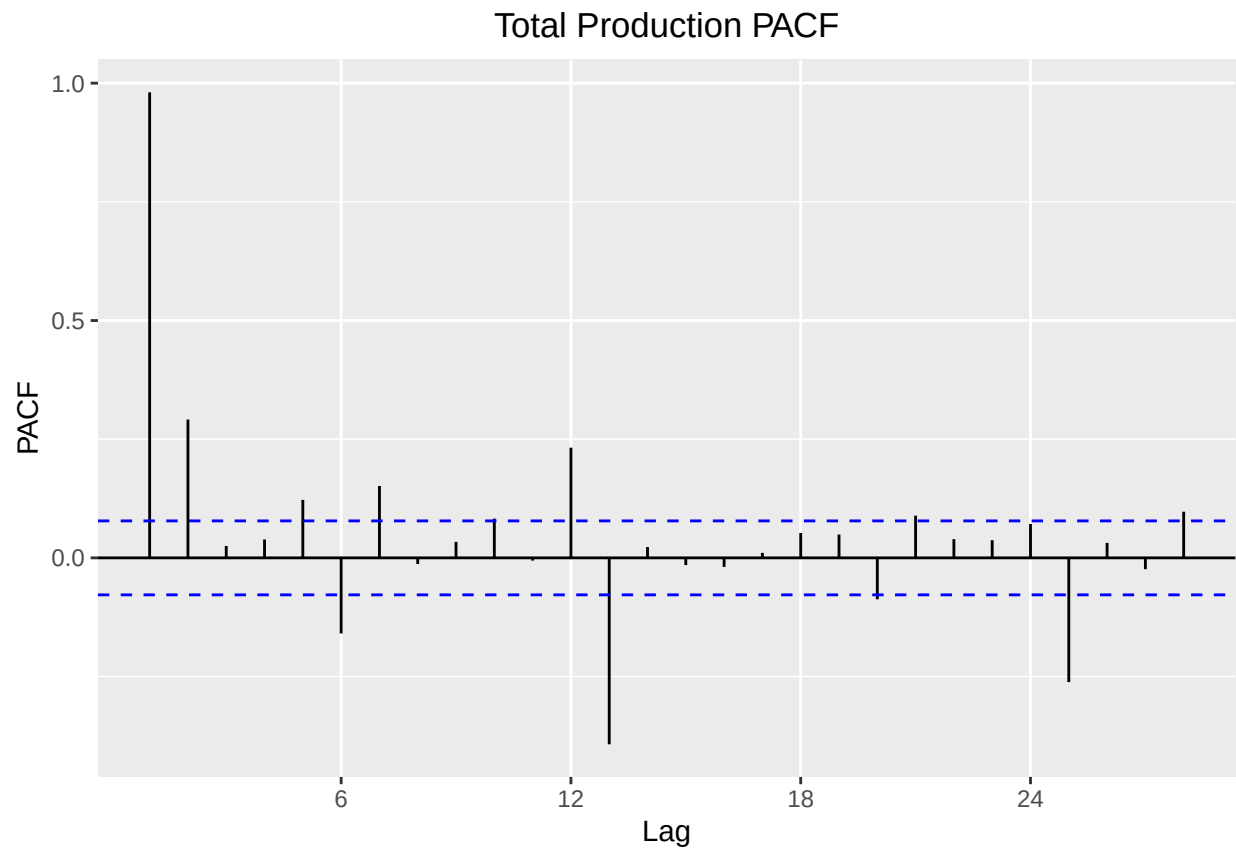
```
#ACF
total_production_acf <- ggAcf(total_production_ts[,2]) +
  labs(title = "Total Production ACF") +
  theme(plot.title = element_text(hjust = 0.5))
total_production_acf
```



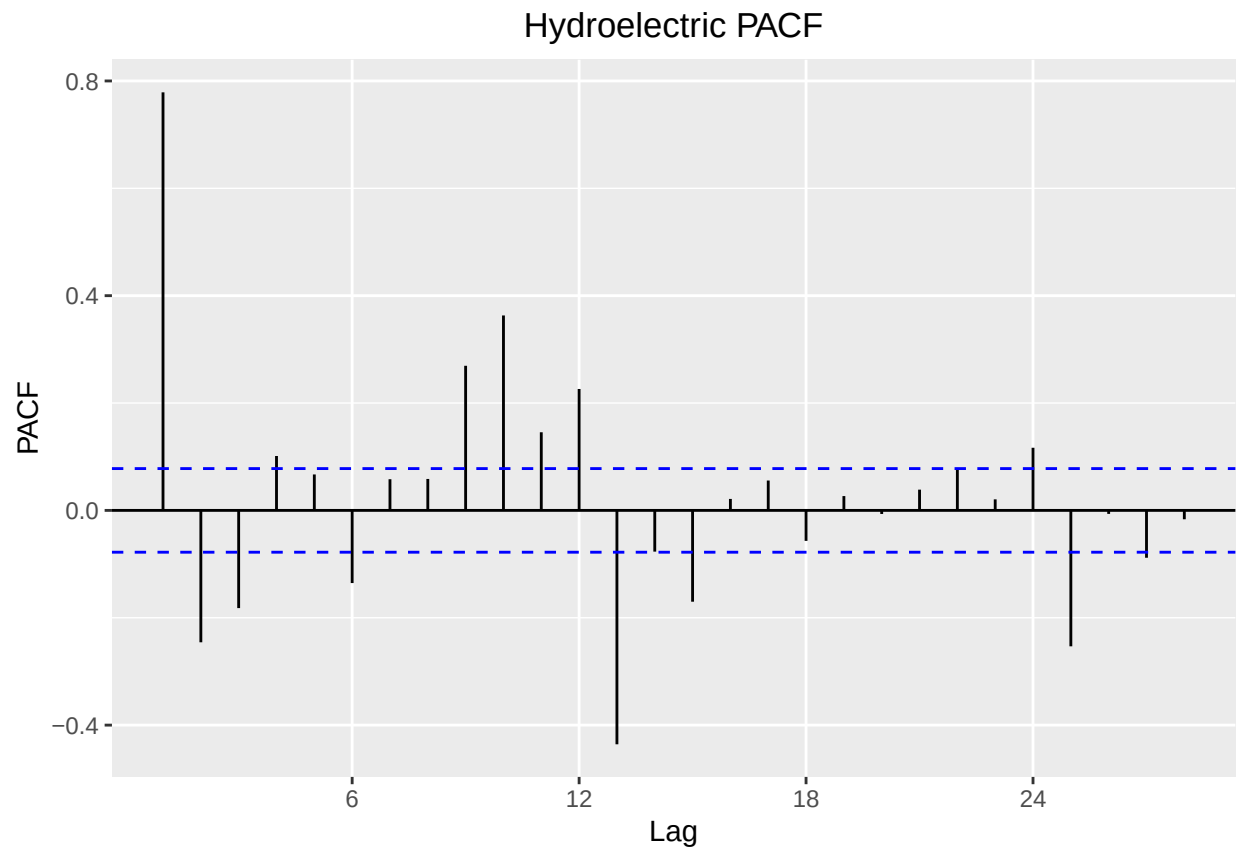
```
hydro_consumption_acf <- ggAcf(hydro_consumption_ts[,2]) +  
  labs(title = "Hydroelectric ACF") +  
  theme(plot.title = element_text(hjust = 0.5))  
hydro_consumption_acf
```



```
#PACF
total_production_pacf <- ggPacf(total_production_ts[,2]) +
  labs(title = "Total Production PACF") +
  theme(plot.title = element_text(hjust = 0.5))
total_production_pacf
```

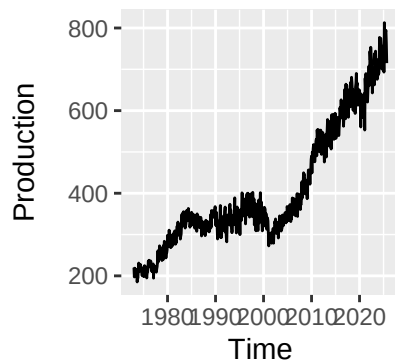


```
hydro_consumption_pacf <- ggPacf(hydro_consumption_ts[,2]) +  
  labs(title = "Hydroelectric PACF") +  
  theme(plot.title = element_text(hjust = 0.5))  
hydro_consumption_pacf
```

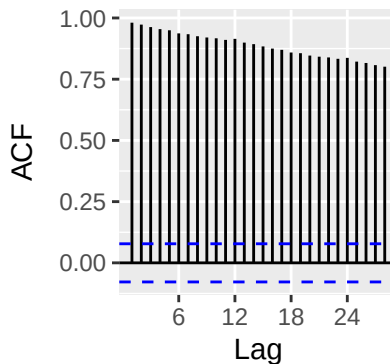


```
#combined  
combined <- plot_grid(total_plot, total_production_acf, total_production_pacf, hydro_plot, hydro_consumption,   
                      nrow = 2, ncol = 3, align = "hv")  
combined
```

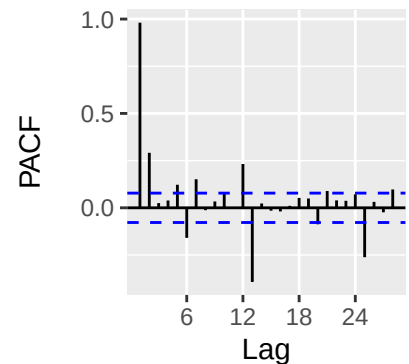
Total Renewable Energy Production



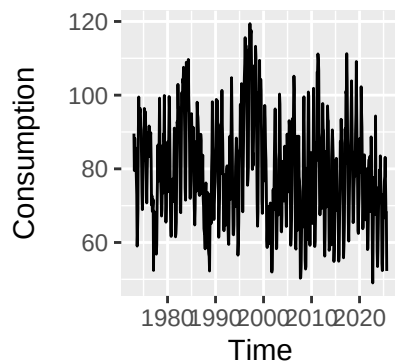
Total Production ACF



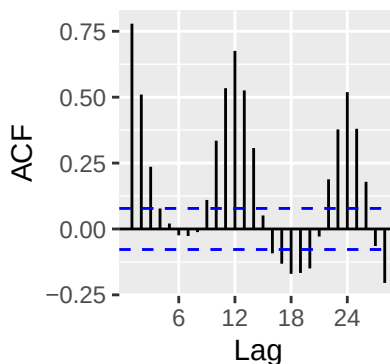
Total Production PAC



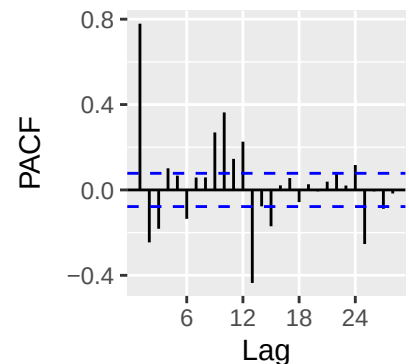
Hydroelectric Energy Consumption



Hydroelectric ACF



Hydroelectric PACF



Q2

From the plot in Q1, do the series Total Renewable Energy Production and Hydroelectric Power Consumption appear to have a trend? If yes, what kind of trend?

Answer: Total Renewable Energy Production has a clear upward trend. This is also supported by the ACF plot where ACF stays very high after many lags. Hydroelectric Power Consumption doesn't appear to have a trend, but it shows seasonality component as shown in ACF plot.

Q3

Use the `lm()` function to fit a linear trend to the two time series. Ask R to print the summary of the regression. Interpret the regression output, i.e., slope and intercept. Save the regression coefficients for further analysis.

```
#create a vector t
nobs_renewable <- nrow(total_production_ts)
t_renewable <- c(1:nobs_renewable)

#linear model for renewable production
total_lm <- lm(total_production_ts[,2] ~ t_renewable)
summary(total_lm)
```

##

```
## Call:
## lm(formula = total_production_ts[, 2] ~ t_renewable)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -154.81  -39.55   12.52   41.49  171.15
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 171.44868    5.11085   33.55  <2e-16 ***
## t_renewable   0.74999    0.01397   53.69  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 64.22 on 631 degrees of freedom
## Multiple R-squared:  0.8204, Adjusted R-squared:  0.8201
## F-statistic: 2883 on 1 and 631 DF, p-value: < 2.2e-16
```

```
beta0_renewable <- as.numeric(total_lm$coefficients[1])
beta1_renewable <- as.numeric(total_lm$coefficients[2])
```

```
#linear model for hydroelectric consumption
```

```
nobs_hydro <- nrow(hydro_consumption_ts)
```

```
t_hydro <- c(1:nobs_hydro)
```

```
hydro_lm <- lm(hydro_consumption_ts[,2] ~ t_hydro)
summary(hydro_lm)
```

```
##
## Call:
## lm(formula = hydro_consumption_ts[, 2] ~ t_hydro)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -30.190 -10.214  -0.715    8.909   39.723
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 83.223802    1.110552   74.939  < 2e-16 ***
## t_hydro      -0.012199    0.003035   -4.019 6.55e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.95 on 631 degrees of freedom
## Multiple R-squared:  0.02496, Adjusted R-squared:  0.02342
## F-statistic: 16.15 on 1 and 631 DF, p-value: 6.547e-05
```

```
beta0_hydro <- as.numeric(hydro_lm$coefficients[1])
beta1_hydro <- as.numeric(hydro_lm$coefficients[2])
```

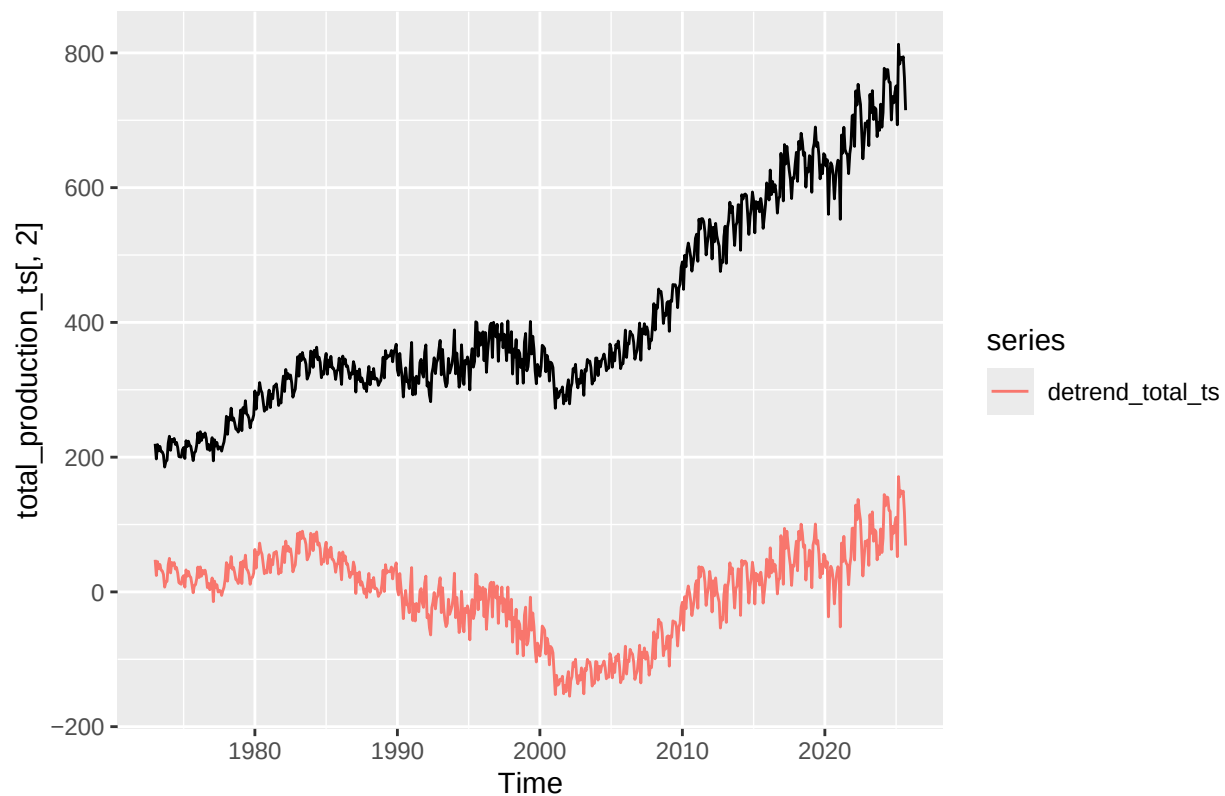
Q4

Use the regression coefficients to detrend each series (subtract fitted linear trend). Plot detrended series and compare with the original time series from Q1. Describe what changed.

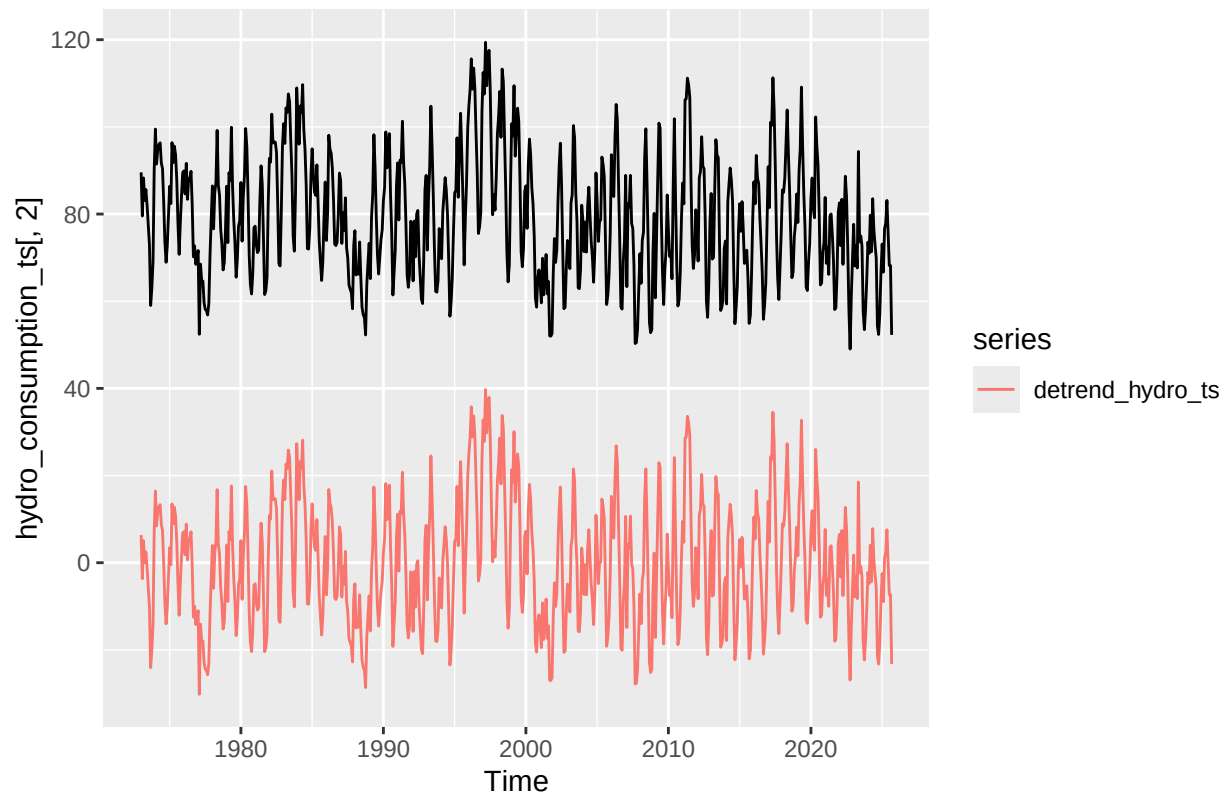
```
#detrend
detrrend_total <- total_production_ts[,2] - (beta0_renewable + beta1_renewable * t_renewable)
detrrend_hydro <- hydro_consumption_ts[,2] - (beta0_hydro + beta1_hydro * t_hydro)

#detrend ts
detrrend_total_ts <- ts(detrrend_total, frequency = 12, start = c(1973,1))
detrrend_hydro_ts <- ts(detrrend_hydro, frequency = 12, start = c(1973,1))

#graph
detrrend_total_plot <- autoplot(total_production_ts[,2]) +
  autolayer(detrrend_total_ts)
detrrend_total_plot
```



```
detrrend_hydro_plot <- autoplot(hydro_consumption_ts[,2]) +
  autolayer(detrrend_hydro_ts)
detrrend_hydro_plot
```



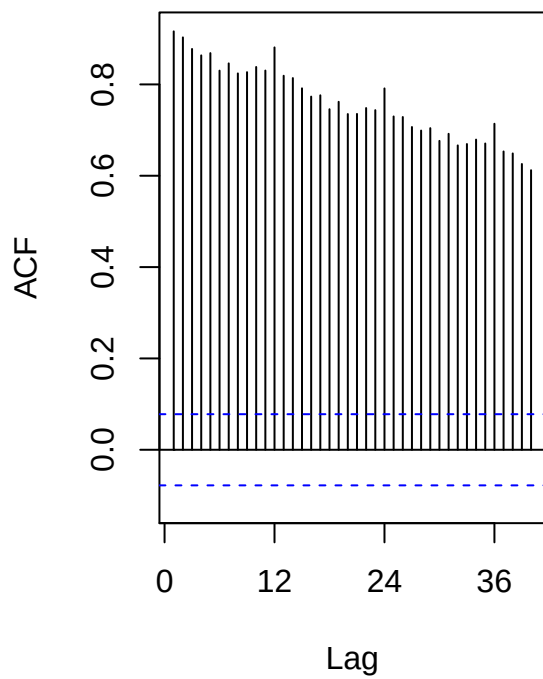
Answer: The original plot for total renewable production shows a upward trend with production after 2000 increasing really fast. But, after removing trend component, the plot changes by having a decreasing trend in the middle, and the overall trend no longer seems increasing. For hydroelectric consumption, the trends look the same but detrended plot shows a lower magnitude compared to the original.

Q5

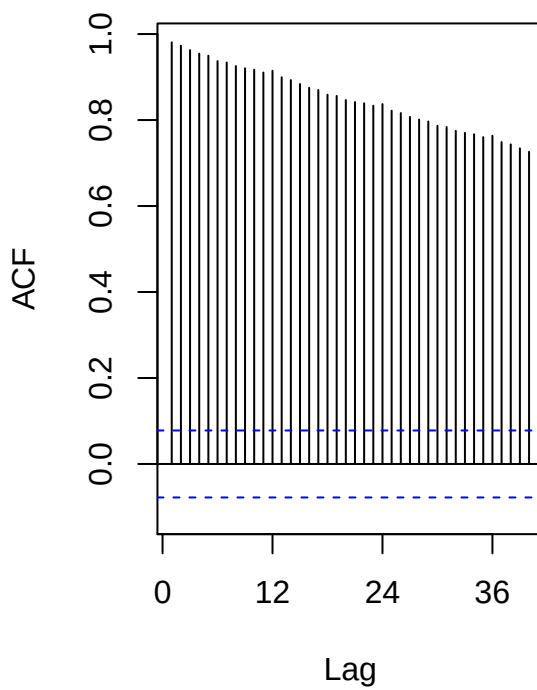
Plot ACF and PACF for the detrended series and compare with the plots from Q1. You may use `plot_grid()` again to get them side by side to make it easier to compare. Did the plots change? How?

```
#total renewable production ACF comparison
detrend_total_acf <- par(mfrow = c(1,2))
Acf(detrend_total_ts, lag.max = 40)
Acf(total_production_ts[,2], lag.max = 40)
```

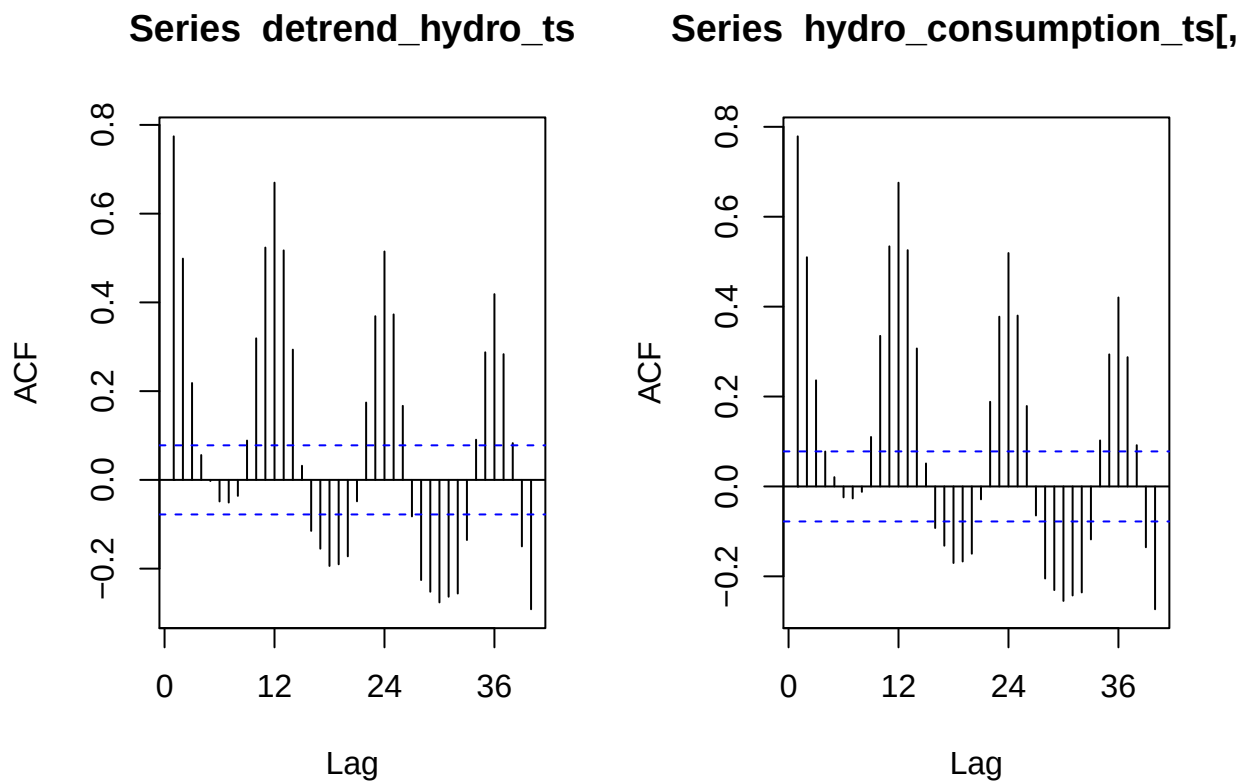
Series detrend_total_ts



Series total_production_ts[, 2]

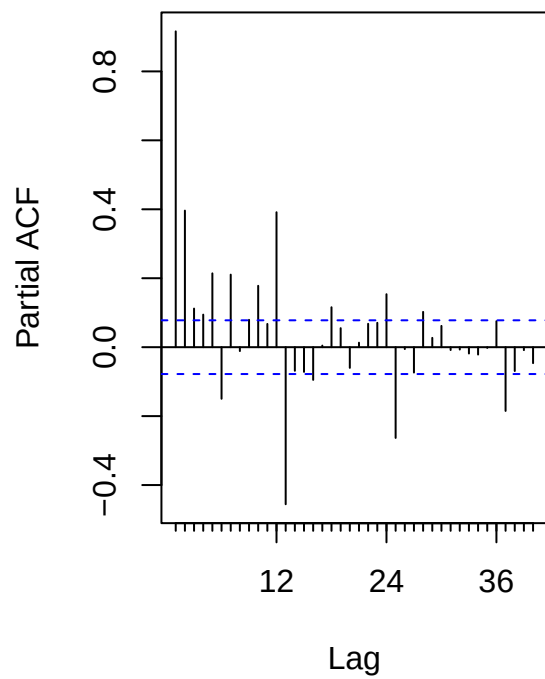


```
#hydro consumption ACF comparison  
detrend_hydro_acf <- par(mfrow = c(1,2))  
Acf(detrend_hydro_ts, lag.max = 40)  
Acf(hydro_consumption_ts[,2], lag.max = 40)
```

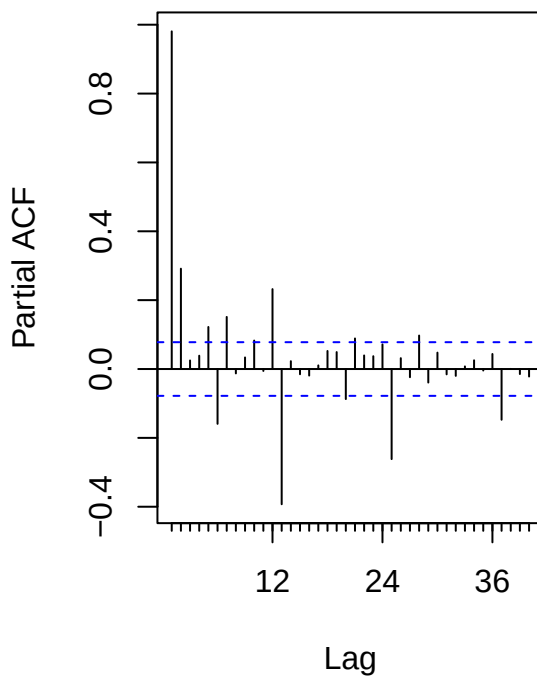


```
#total renewable production PACF comparison
detrend_total_pacf <- par(mfrow = c(1,2))
Pacf(detrend_total_ts, lag.max = 40)
Pacf(total_production_ts[,2], lag.max = 40)
```

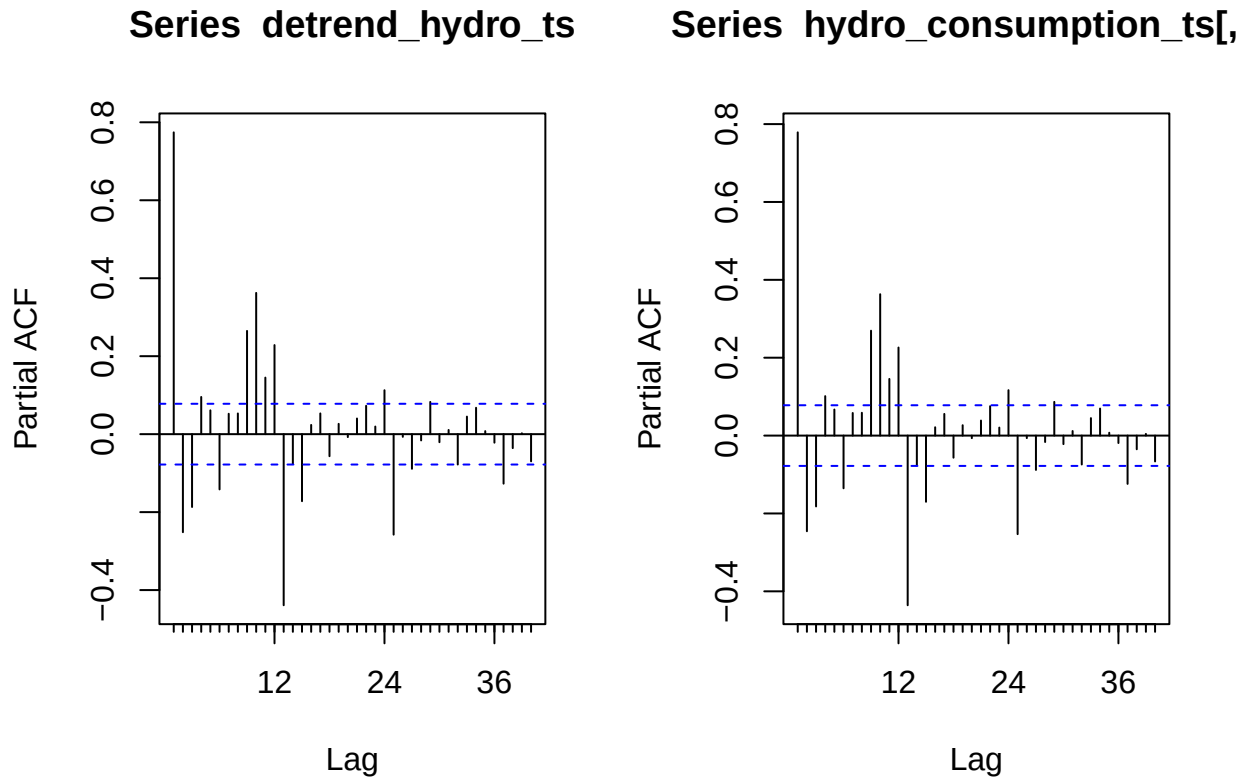
Series detrend_total_ts



Series total_production_ts[, 2]



```
#hydro consumption PACF comparison
detrend_hydro_pacf <- par(mfrow = c(1,2))
Pacf(detrend_hydro_ts, lag.max = 40)
Pacf(hydro_consumption_ts[,2], lag.max = 40)
```



Answer: Total renewable production's ACF plot changed a little after detrending. Before (on the right) shows a more gradual declining trend while now (on the left) has some spikes while an overall declining trend. For hydroelectric consumption, before (on the right) is the same as now (on the left). On the other hand, total renewable production's PACF looks very similar before and after, with detrended version (on the left) having lower magnitude. Hydroelectric's PACF looks the same before and after.

Seasonal Component

Set aside the detrended series and consider the original series again from Q1 to answer Q6 to Q8.

Q6

Just by looking at the time series and the acf plots, do the series seem to have a seasonal trend? No need to run any code to answer your question. Just type in you answer below.

Answer: Total renewable production does not seem having a seasonal trend because after removing trend component, time series plot changed with some flunctuations. Its ACF plot shows strong positive persistance after many lags. This means total renewable production has a trend but could not simply explained as a linear trend. On the other hand, hydro consumption shows the same plot with lower magnitude after removing trend component. ACF plot shows strong seasonal trend by having repeating spikes at regular intervals.

Q7

Use function `lm()` to fit a seasonal means model (i.e. using the seasonal dummies) to the two time series. Ask R to print the summary of the regression. Interpret the regression output. From the results, which series have a seasonal trend? Do the results match you answer to Q6?

```
#create dummies
dummies_total <- seasonaldummy(total_production_ts[,2])
dummies_hydro <- seasonaldummy(hydro_consumption_ts[,2])

#seasonal mean model
total_means_model <- lm(total_production_ts[,2] ~ dummies_total)
summary(total_means_model)
```

```
##
## Call:
## lm(formula = total_production_ts[, 2] ~ dummies_total)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -213.33  -97.36  -59.88   121.55   389.62
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    417.265     21.096   19.779  <2e-16 ***
## dummies_totalJan     2.090     29.693    0.070   0.944
## dummies_totalFeb   -34.524     29.693   -1.163   0.245
## dummies_totalMar     5.956     29.693    0.201   0.841
## dummies_totalApr    -6.900     29.693   -0.232   0.816
## dummies_totalMay     8.162     29.693    0.275   0.784
## dummies_totalJun    -2.231     29.693   -0.075   0.940
## dummies_totalJul     3.864     29.693    0.130   0.897
## dummies_totalAug    -3.978     29.693   -0.134   0.893
## dummies_totalSep   -29.033     29.693   -0.978   0.329
## dummies_totalOct   -19.937     29.834   -0.668   0.504
## dummies_totalNov   -20.617     29.834   -0.691   0.490
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 152.1 on 621 degrees of freedom
## Multiple R-squared:  0.008243, Adjusted R-squared: -0.009324
## F-statistic: 0.4692 on 11 and 621 DF, p-value: 0.9223
```

```
hydro_means_model <- lm(hydro_consumption_ts[,2] ~ dummies_total)
summary(hydro_means_model)
```

```
##
## Call:
## lm(formula = hydro_consumption_ts[, 2] ~ dummies_total)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -30.895  -6.368  -0.595    6.213   32.557
```

```
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)    79.724     1.436  55.511 < 2e-16 ***
## dummies_totalJan     4.951     2.021   2.449 0.014591 *
## dummies_totalFeb    -2.415     2.021  -1.195 0.232608
## dummies_totalMar     7.116     2.021   3.520 0.000463 ***
## dummies_totalApr     5.614     2.021   2.777 0.005649 **
## dummies_totalMay    14.080     2.021   6.965 8.38e-12 ***
## dummies_totalJun    10.780     2.021   5.333 1.36e-07 ***
## dummies_totalJul     4.003     2.021   1.980 0.048091 *
## dummies_totalAug    -5.320     2.021  -2.632 0.008710 **
## dummies_totalSep   -16.598     2.021  -8.211 1.28e-15 ***
## dummies_totalOct   -16.329     2.031  -8.040 4.56e-15 ***
## dummies_totalNov   -10.782     2.031  -5.308 1.54e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.36 on 621 degrees of freedom
## Multiple R-squared:  0.4714, Adjusted R-squared:  0.4621
## F-statistic: 50.35 on 11 and 621 DF,  p-value: < 2.2e-16
```

Answer: For total renewable production, the P-value is 0.9223 which is larger than 0.05 which means it does not have seasonal component. For hydro consumption, the p-value is very small and smaller than 0.05 which means it has seasonal component. This result matches my answer to Q6.

Q8

Use the regression coefficients from Q7 to deseason the series. Plot the deseason series and compare with the plots from part Q1. Did anything change?

```
#store coefficient
int_total_mean_model <- total_means_model$coefficients[1]
coeff_total_mean_model <- total_means_model$coefficients[2:12]

int_hydro_mean_model <- hydro_means_model$coefficients[1]
coeff_hydro_mean_model <- hydro_means_model$coefficients[2:12]

#compute seas component
season_total <- array(0,nobs_renewable)
for(i in 1:nobs_renewable) {
  season_total[i] <- int_total_mean_model + coeff_total_mean_model %*% dummies_total[i,]
}

season_hydro <- array(0,nobs_hydro)
for(i in 1:nobs_hydro) {
  season_hydro[i] <- int_hydro_mean_model + coeff_hydro_mean_model %*% dummies_hydro[i,]
}

#removing seas component from series
```

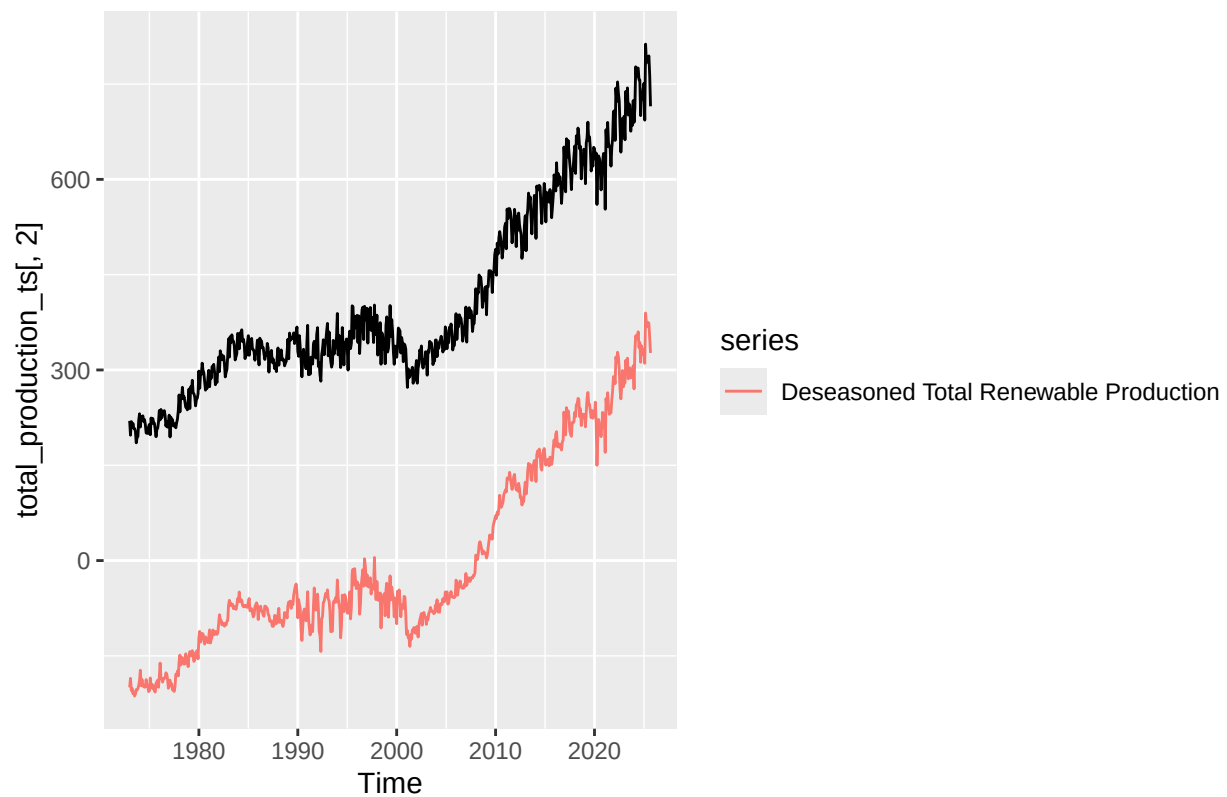
```

deseason_total <- total_renewable_production - season_total
deseason_hydro <- hydro_consumption - season_hydro

#convert to ts obj
deseason_total_ts <- ts(deseason_total, start = c(1973,1), frequency = 12)
deseason_hydro_ts <- ts(deseason_hydro, start = c(1973,1), frequency = 12)

#plots
deseason_total_plot <- autoplot(total_production_ts[,2]) +
  autolayer(deseason_total_ts[,2], series = "Deseasoned Total Renewable Production")
deseason_total_plot

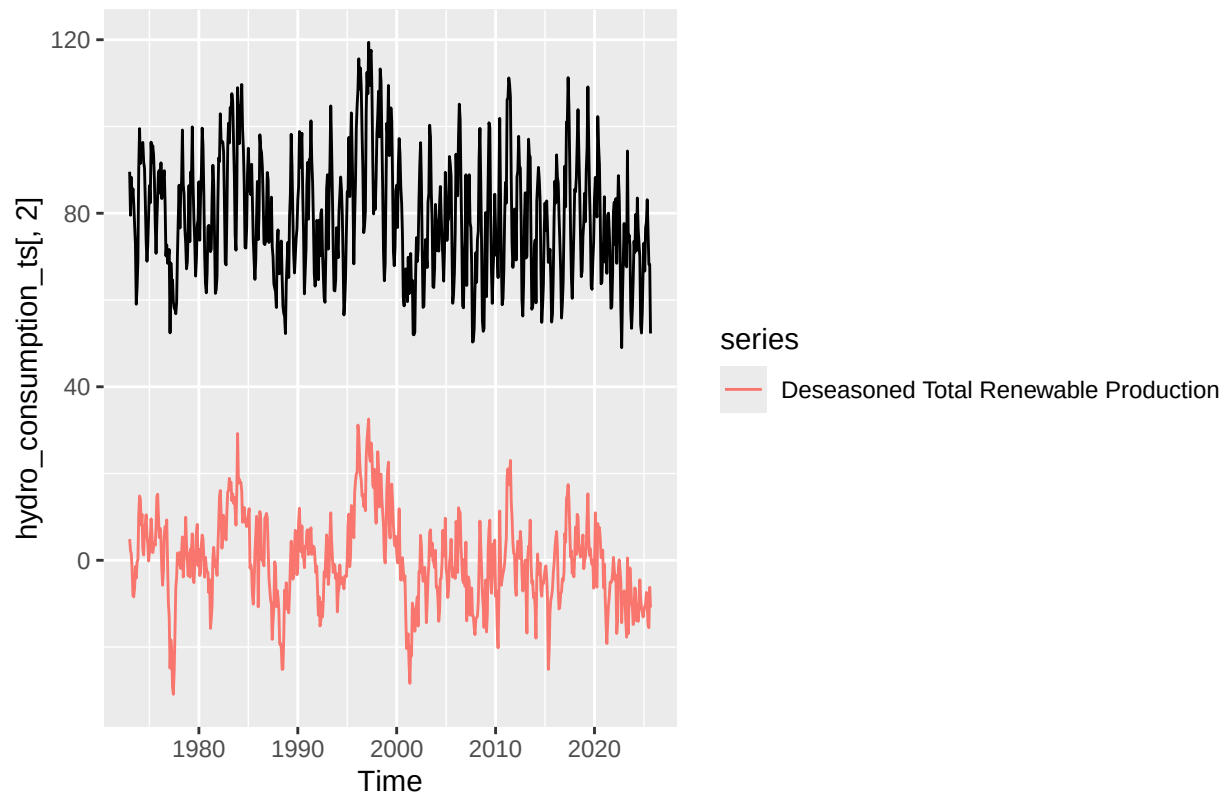
```



```

deseason_hydro_plot <- autoplot(hydro_consumption_ts[,2]) +
  autolayer(deseason_hydro_ts[,2], series = "Deseasoned Total Renewable Production")
deseason_hydro_plot

```



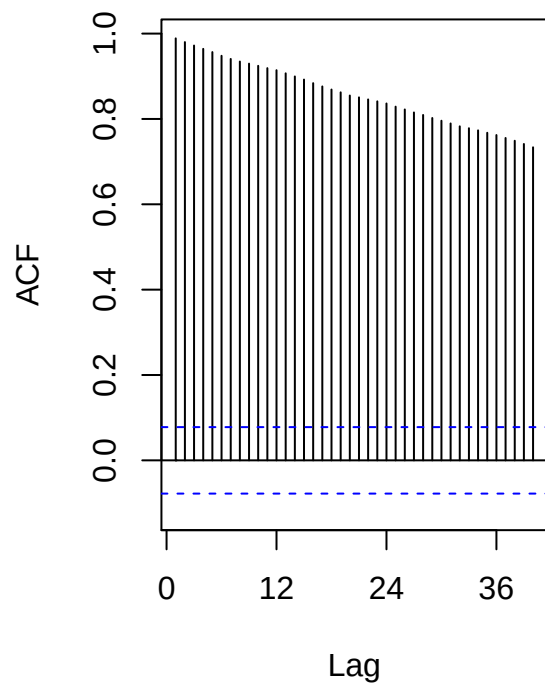
Answer: For total renewable production, the trend looks similar compared to the original with lower magnitude. For hydro consumption, new plot has some changes compared to the original plot which indicate it has some seasonal trend being removed.

Q9

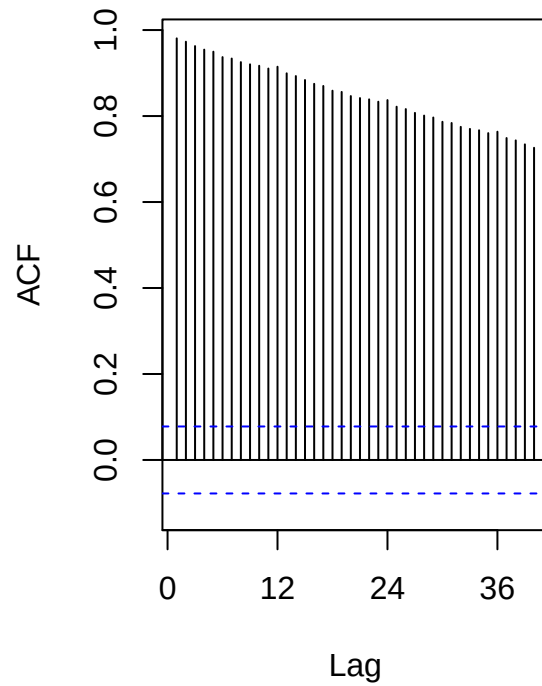
Plot ACF and PACF for the deseason series and compare with the plots from Q1. You may use `plot_grid()` again to get them side by side. Did the plots change? How?

```
#ACF
deseason_total_acf <- par(mfrow = c(1,2))
  Acf(deseason_total_ts[,2], lag.max = 40)
  Acf(total_production_ts[,2], lag.max = 40)
```

Series deseason_total_ts[, 2]

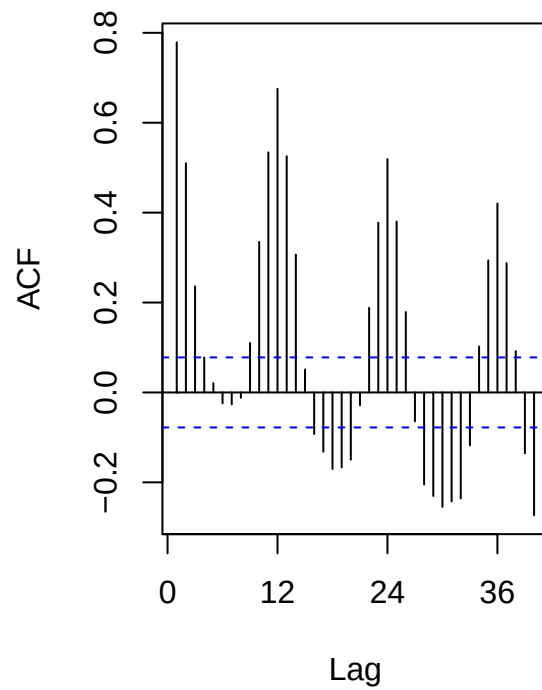
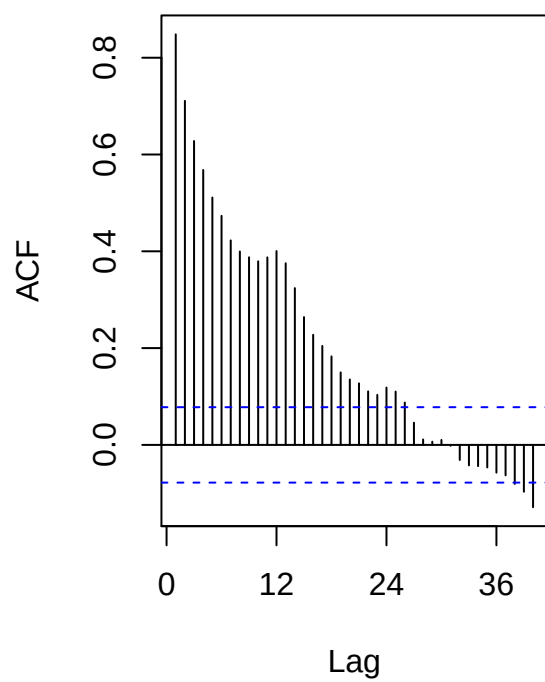


Series total_production_ts[, 2]



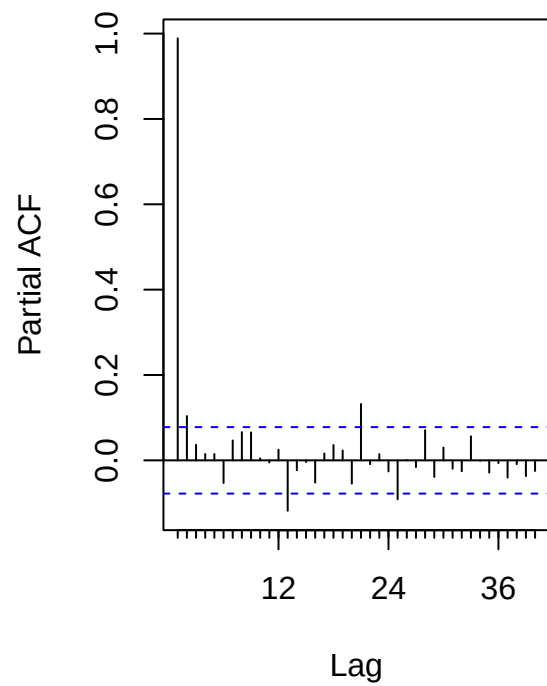
```
deseason_hydro_acf <- par(mfrow = c(1,2))  
Acf(deseason_hydro_ts[,2], lag.max = 40)  
Acf(hydro_consumption_ts[,2], lag.max = 40)
```

Series deseason_hydro_ts[, 2] Series hydro_consumption_ts[, 2]

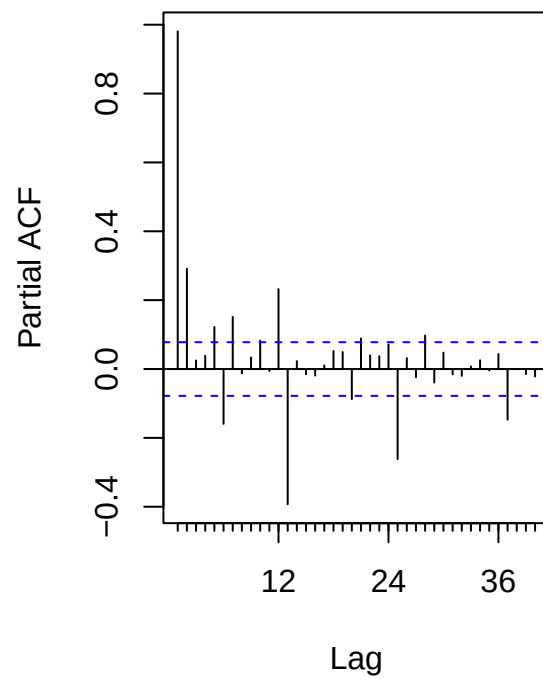


```
#PACF
deseason_total_pacf <- par(mfrow = c(1,2))
Pacf(deseason_total_ts[,2], lag.max = 40)
Pacf(total_production_ts[,2], lag.max = 40)
```

Series deseason_total_ts[, 2]

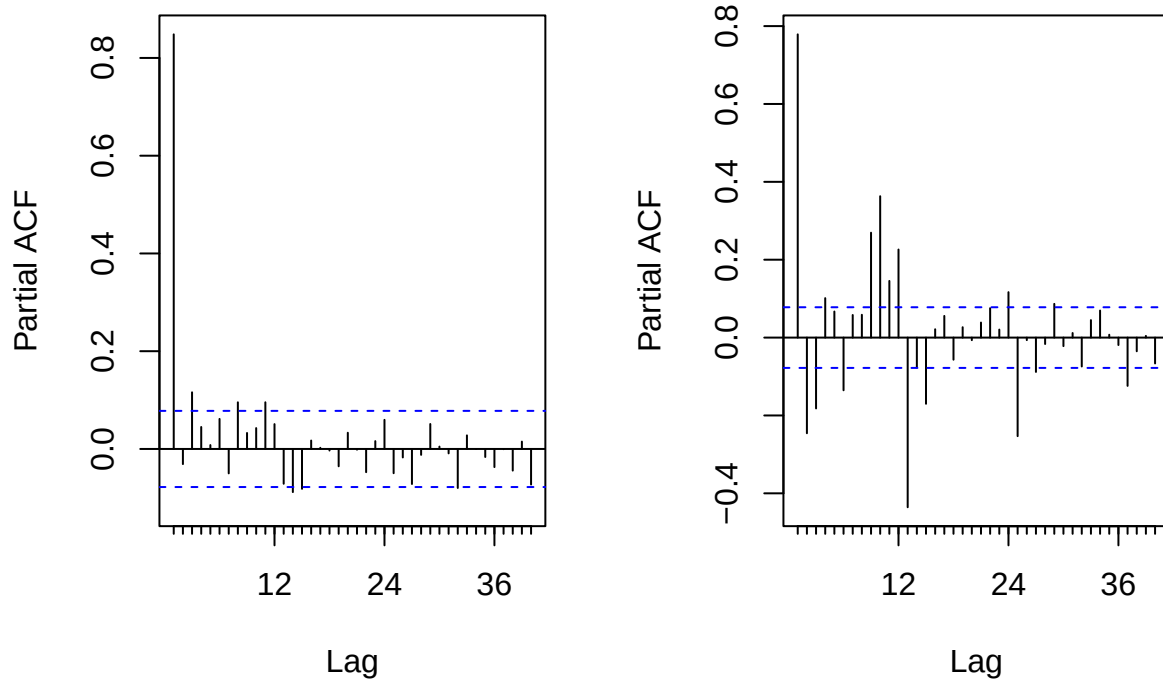


Series total_production_ts[, 2]



```
deseason_hydro_pacf <- par(mfrow = c(1,2))
Pacf(deseason_hydro_ts[,2], lag.max = 40)
Pacf(hydro_consumption_ts[,2], lag.max = 40)
```

Series deseason_hydro_ts[, 2] Series hydro_consumption_ts[,



Answer: ACF for total renewable production barely changed after removing seasonal component while PACF also shows similar pattern with marginal changes. For hydro consumption, ACF changed after removing seasonal component. Unlike original plot with repeating patterns, the new plot shows declining trend all the way to 0 then to negative. PACF also changed with less negative and positive magnitude.